Course: [BCS-301] Data Structures

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: LECTURE NOTES**

Module / Unit: Unit 2 **Faculty Author: Dr. R. K. Sharma (Associate P**

Unit 2 Lecture Notes: Linked Lists, Binary Trees & Balanced Trees

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1. Singly, Doubly, and Circular Linked Lists

- Dynamic Memory Allocation: Pointer linking, node structure, head and tail pointers.
- Operations: Insertion and deletion at beginning, intermediate, and end positions.
- Advantages over arrays: Dynamic size flexibility and constant-time node insertion.

2. Binary Trees and Binary Search Trees (BST)

- Tree Terminology: Root, leaf, internal nodes, depth, height, level order.
- Tree Traversals: In-order (Left-Root-Right), Pre-order (Root-Left-Right), Post-order.
- BST Invariant: Left subtree keys < root key < right subtree keys. Search complexity O(h).

3. Self-Balancing Trees: AVL Trees & Rotations

- Balance Factor: BF = Height(Left) - Height(Right), constrained to {-1, 0, +1}.
- Tree Rotations: Single Rotations (LL, RR) and Double Rotations (LR, RL) to restore balance.
- Worst-case Search Guarantee: O(log n) guaranteed search, insert, and delete operations.